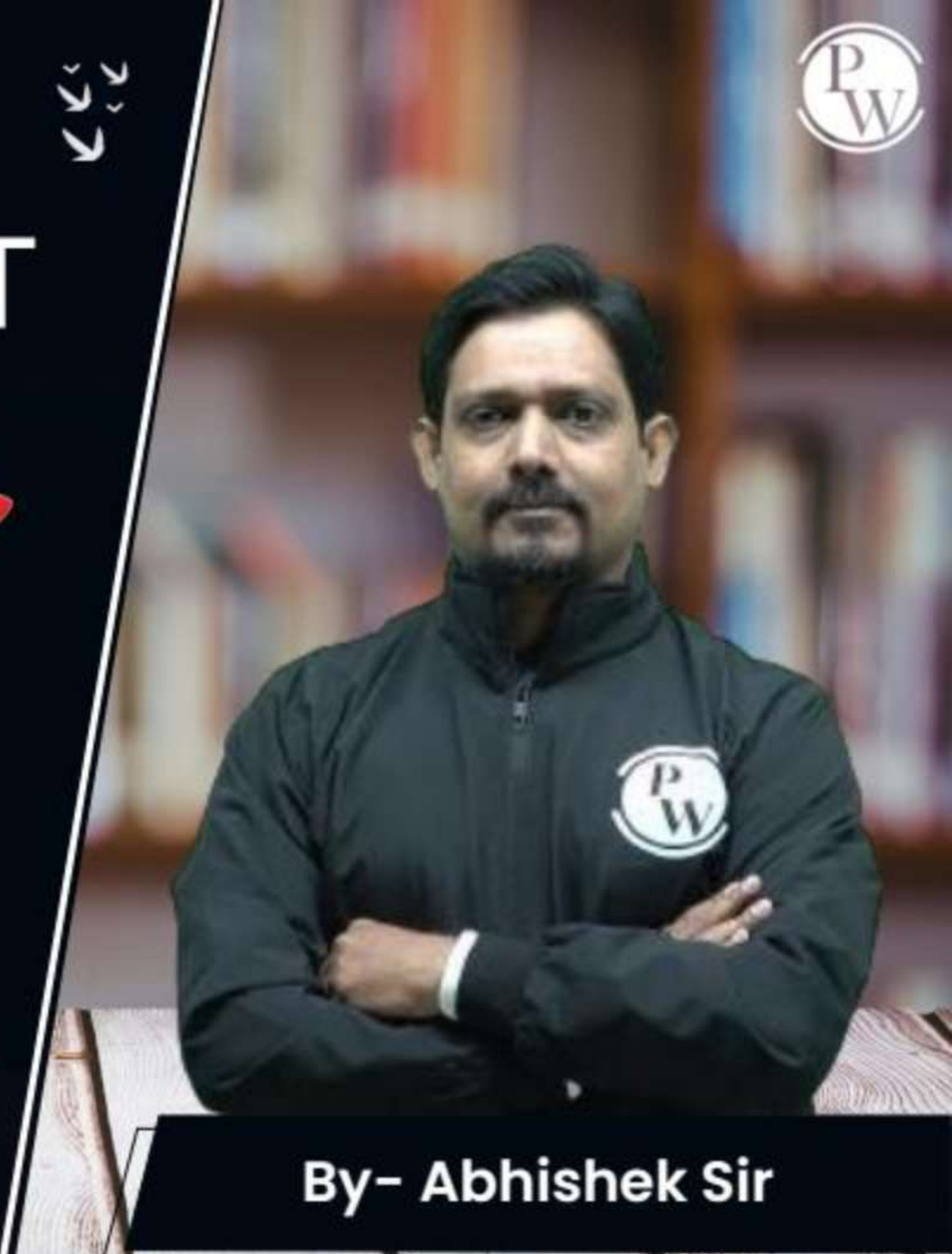
Computer Science & IT



C programming

Function & Storage Class

Lecture No. 04



By- Abhishek Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic
- Topic
- Topic

Recursion

Single Recursion
- Tail
- NonTail

Multiple Recursion

Slide

Topics to be Covered



- Topic — Indirect Recursion
- Topic — Nested
- Topic — TOH (Tower of Hanoi)
- Topic
- Topic

Slide



Recursion

#Q. Consider the following program

```
#include<stdio.h>

int sum(int n) {
  n = 3;
  if (n == 2) return 2;
  return sum(n-1)*n ;
}

int main() {
    printf("%d\n", sum(5) );
    return 0

}
```

A. 120

B. Abnormal termination

C. 24

D. 720

Sum(5)

↓

Sum(2) * 3

↓

Sum(2) * 3

⋮

Slide



Recursion

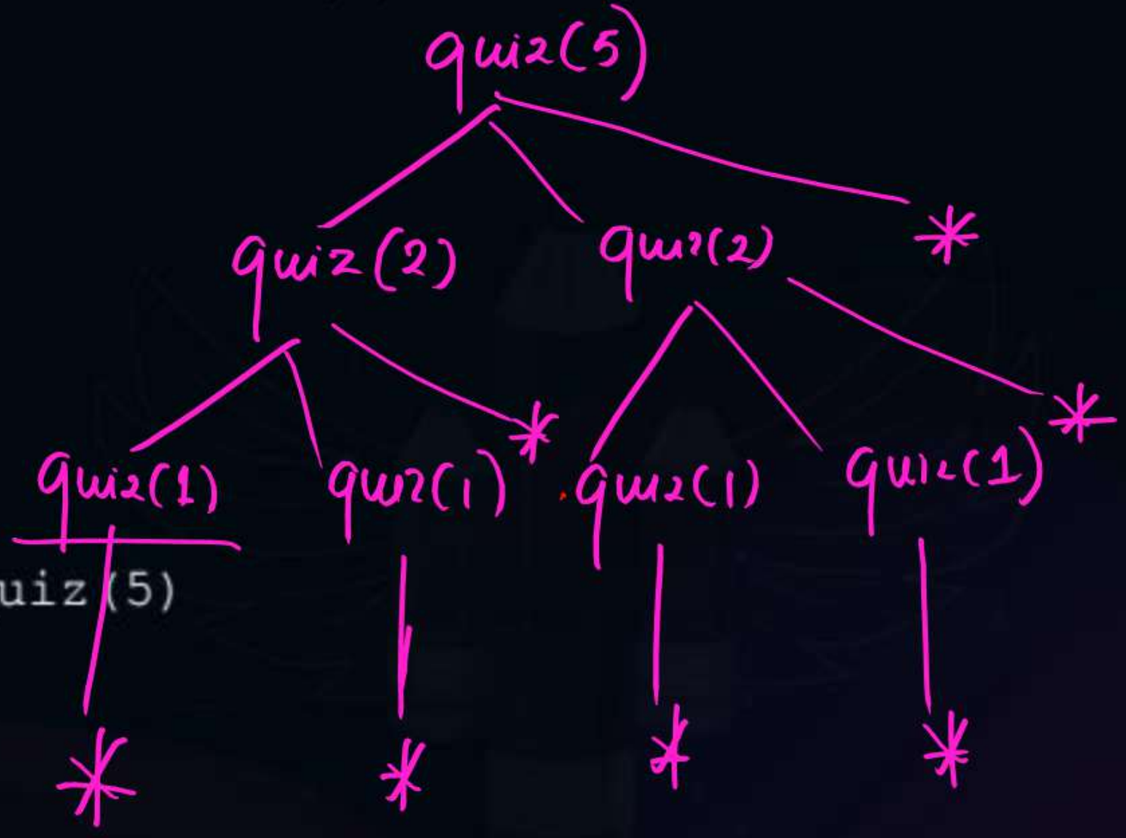
#Q. Consider the following program

```
void quiz(int i)
    {
        if (i > 1)
        {
            quiz(i / 2);
            quiz(i / 2);
        }
        printf("*");
    }
```

What value will be returned by the call quiz(5)

No. of star printed

(A) 8
(B) 7
(C) 3
(D) 2



Slide



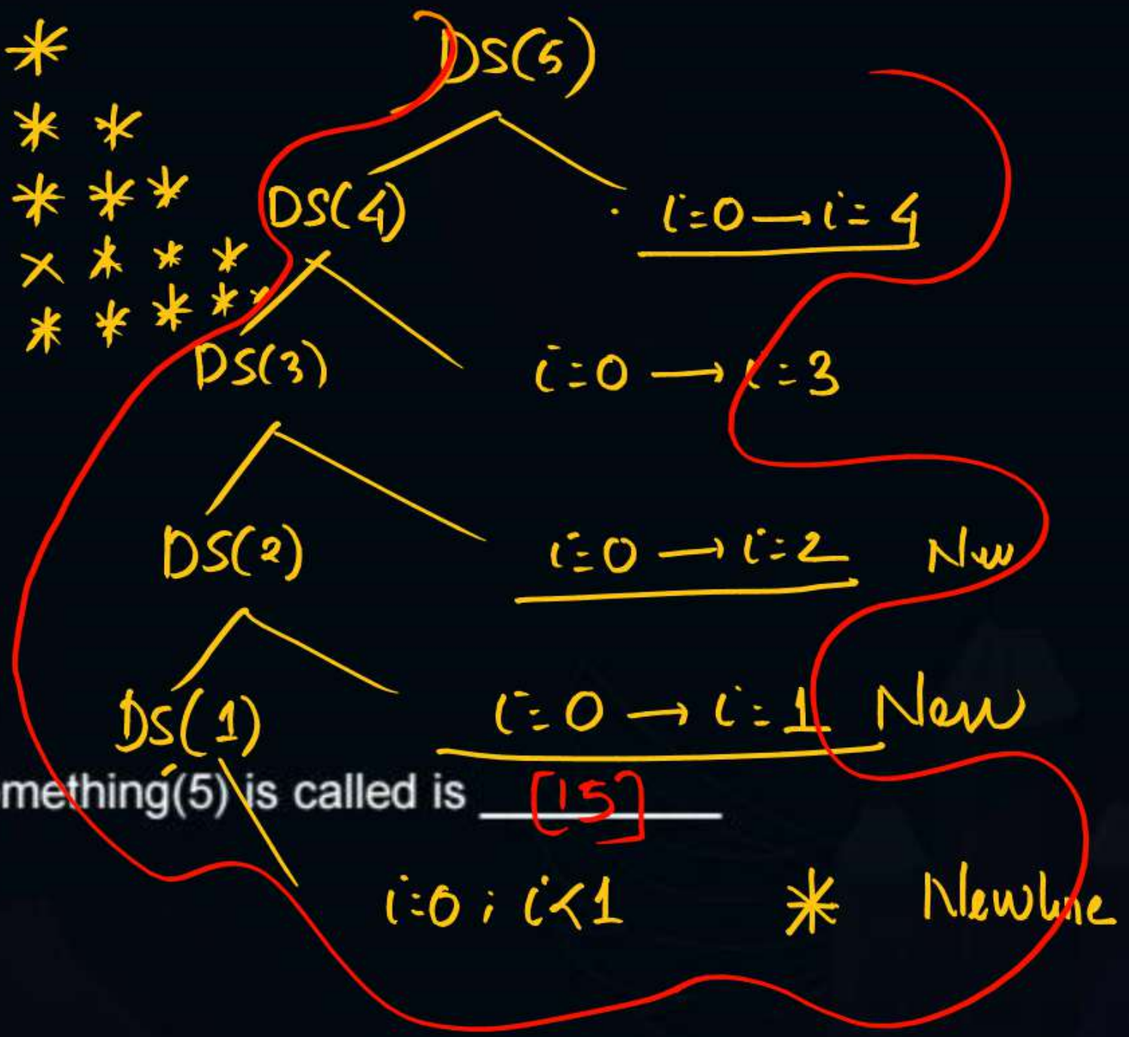
Question

#Q. Consider the following program

```
*
* *
* * *
* * * *
* * * * *
```

```
void Dosomething(int n) {
        if (n > 1)
                Dosomething(n-1);
        for (int i = 0; i < n; i++)
                printf("*");
        printf("\n");
}
```

The number of stars will print if the Dosomething(5) is called is _____



Slide



Question

#Q Consider the C function given below

```
int f(int j){
static int i = 50;
int k;
if (i == j){
        printf("something");
        k = f(i);
        return 0;
}
else
        return 0;
}
```

Which one of the following is TRUE?

(A) The function returns 0 for all values of j.

(B) The function prints the string something for all values of j.

(C) The function returns 0 when j = 50.

(D) The function will exhaust the runtime stack or run into an infinite loop when j = 50.

Consider the following program:

```
int main()
{
   f1();
   f2(2);
   f3();
   return(0);
}
```

```
int f1()
{
  return(1);
}
```

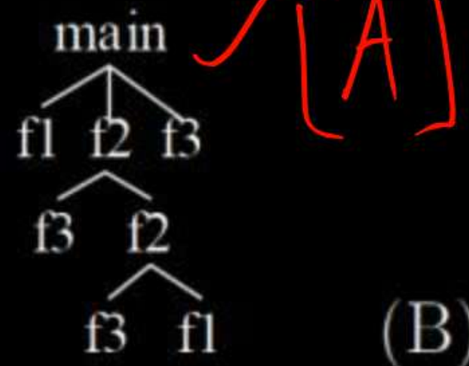
```
int f2(int X)
{
   f3();
  if (X==1)
    return f1();
  else
    return (X*f2(X-1));
}
```

```
int f3()
{
  return(5);
}
```

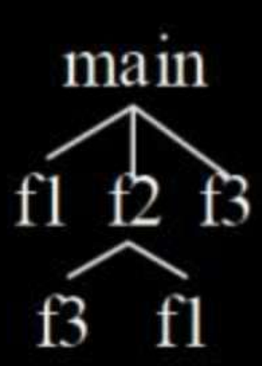
PYQ

Which one of the following options represents the activation tree corresponding to the main function?

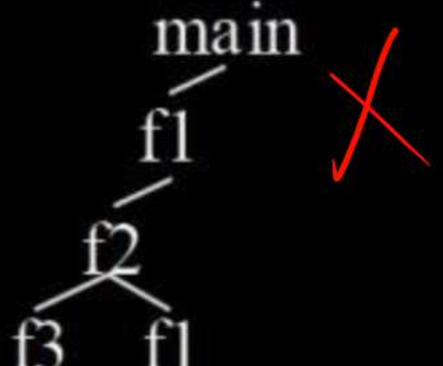
(A)



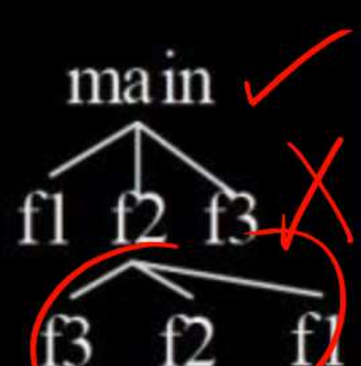
(B)



(C)



(D)



```
{
  return(1);
}

(0);
```

```
{
  f3();
  if (X==1)
    return f1();
  else
    return (X*f2(X-1));
}
```

```
{
  return(5);
}
```

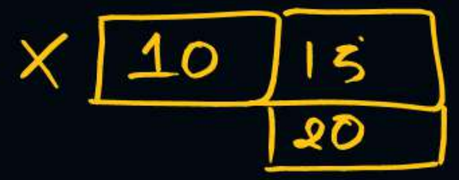
of the following options represents the activation tree corresponding to
unction?



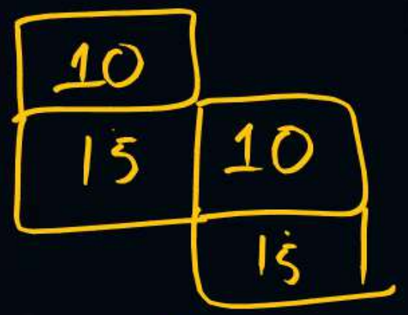
Question

```
#Q # include <stdio.h>
void print1 (void){
    static int x = 10;
    x+= 5;
    printf ("%d\t", x);
}
void print2 (void){
    static int x;
    x= 10;
    x+= 5;
    printf ("%d\t", x);
}
int main (){
print1(); print1(); print2(); print2();
return 0;
}
```

x = x+5



Assignment



[D]

(A) 15 20 25 30

(B) 15 20 15 20

(C) 15 15 15 15

(D) 15 20 15 15 ✓

Initialization

Assignment

15 20 15, 15

Slide



Question

#Q The integer value printed by the ANSI-C program given below is .

```
#include<stdio.h>

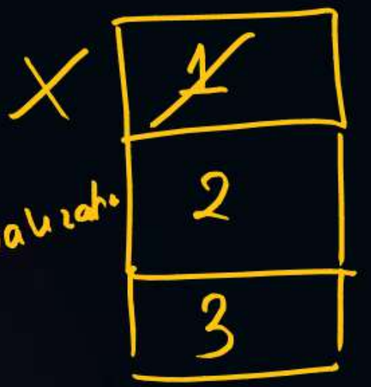
int funcp(){

static int x = 1;

x++;

return x;

}
```



X++ , ++X

Same

```
int main(){

int x,y;

x = funcp();

y = funcp()+x;

printf("%d\n", (x+y));

return 0;

}
```

x = 2

y = 3+2 = 5

5+2 = 7

Slide

#Q Consider the following c-program

```
#include <stdio.h>
int r(){
  static int num=7;
  return num --;
}
int main(){
      for (r();r();r())
            printf("%d",r());
      return 0;
}
```

num | 7 | 6 | 5

1 2 4

3

r()
while(r()){
  printf("%d",r());
r()
}

[A]

Which one of the following values will be displayed on execution of the programs?

(A) 52 (B) 630 (C) 41 (D) 63

Slide

#Q Consider the following c-program

```
#include <stdio.h>
int r(){
  static int num=7;
  return num --;
}
int main(){
      for (r();r();r())
            printf("%d",r());
      return 0;
}
```

1, 2, 4, 3

num | 7 | 6 | 5 | 4 | 3 | 2 | 1

Iteration 1

for(7; 6; 4)

printf(r())

1 2 3

4

5

Iteration 2 ✓

for(; r(); 1)

3

printf("%d" r())

2

[A]

Which one of the following values will be displayed on execution of the programs?

(A) 52 (B) 630 (C) 41 (D) 63

Slide



Recursion

#Q. What value would the following function return for the input x = 97?

(a) 89
(b) 90
(c) 91
(d) 92

```
int fun (int x){
     if x > 100
          return x - 10;
     else
          return  fun(fun (x + 11))
}
```



Recursion

#Q. What value would the following function return for the input x = 97?

```
int fun (int x){
    if x > 100
        return x - 10;
    else
        return fun(fun (x + 11));
}
```

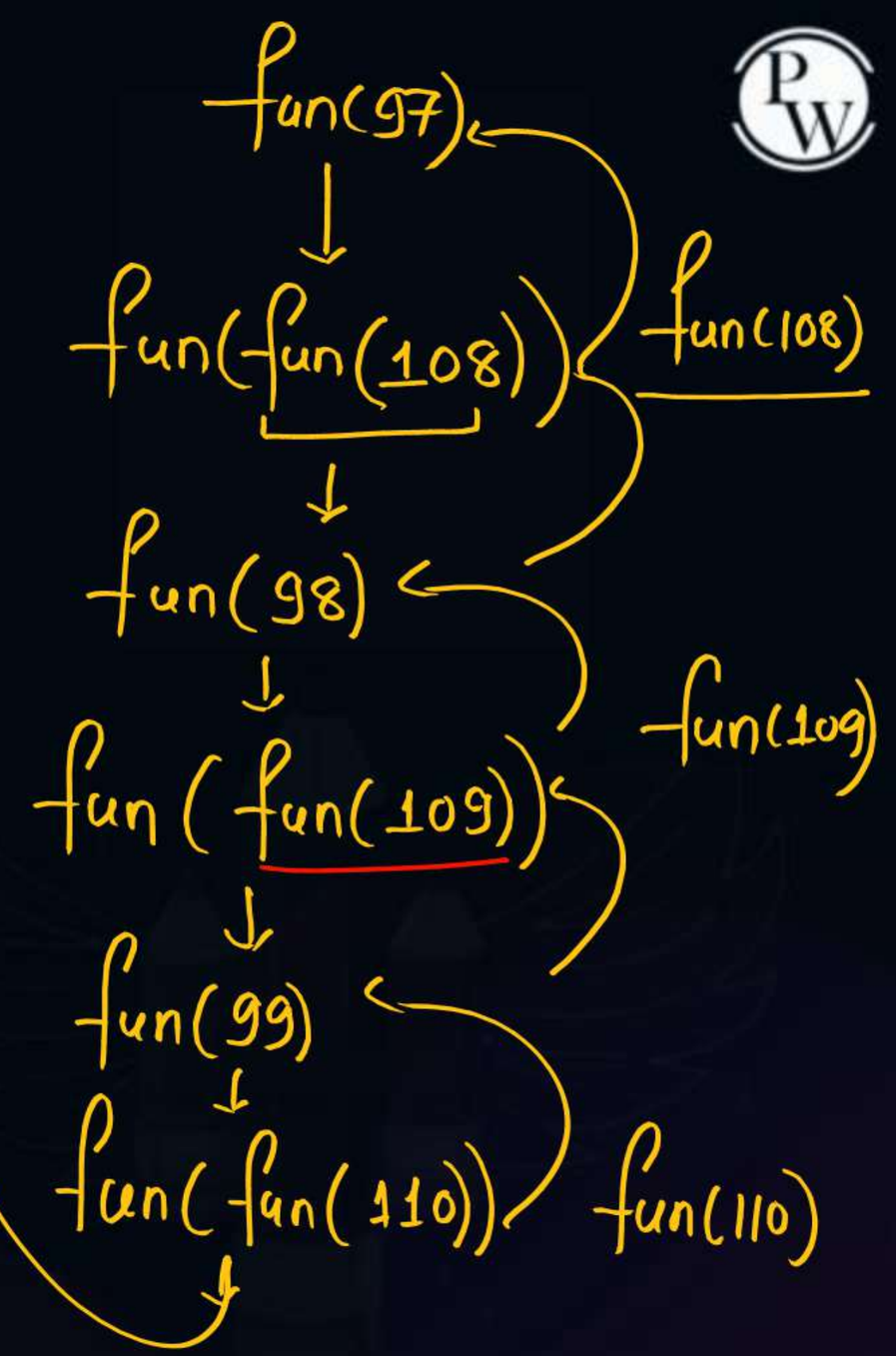
fun(111)

fun(100) → 91

fun(fun(111)) → 91

fun(101) → 91

91



Slide

Nested Recursion

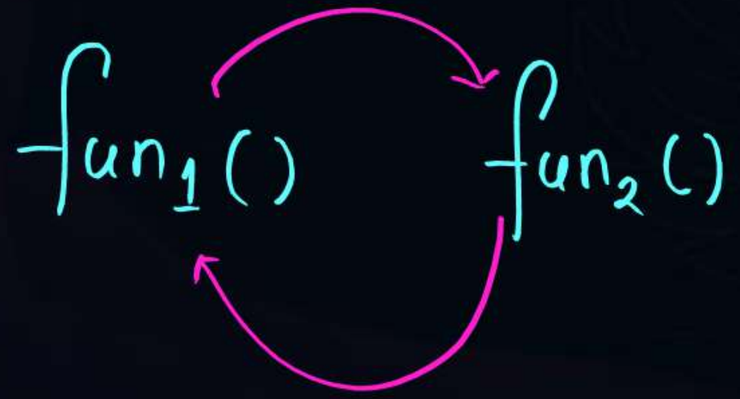
if Recursive call is passed as parameter

to Recursive function then we call that as Nested Recursion.



Indirect Recursion

Indirect Recursion : In Indirect Recursion more than 1 function is present & they call each other in Circular manner.



Slide



Indirect Recursion

3, 1, 2

```
void fun1 (int n) {

if (n = =0 ) return;

printf ("%d" , n);

fun2 (n - 2);

}

int main(){

fun1(3);

}
```

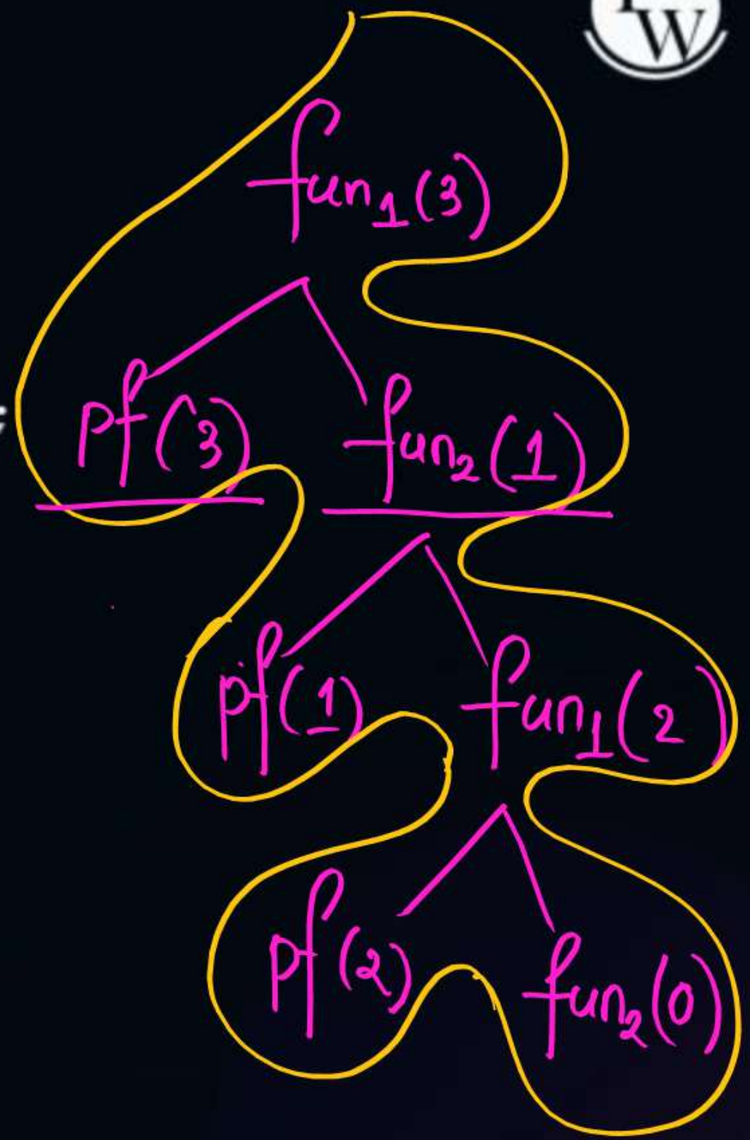
```
void fun2 (int n) {

if (n = = 0) return ;

printf ("%d" , n);

fun1(++n) ;

}
```





Indirect Recursion

```
void fun1 (int n) {

if (n = =0 ) return;

printf ("%d" , n);

fun2 (n - 2);

}

int main(){

fun1(3);

}
```

```
void fun2 (int n) {

if (n = = 0) return ;

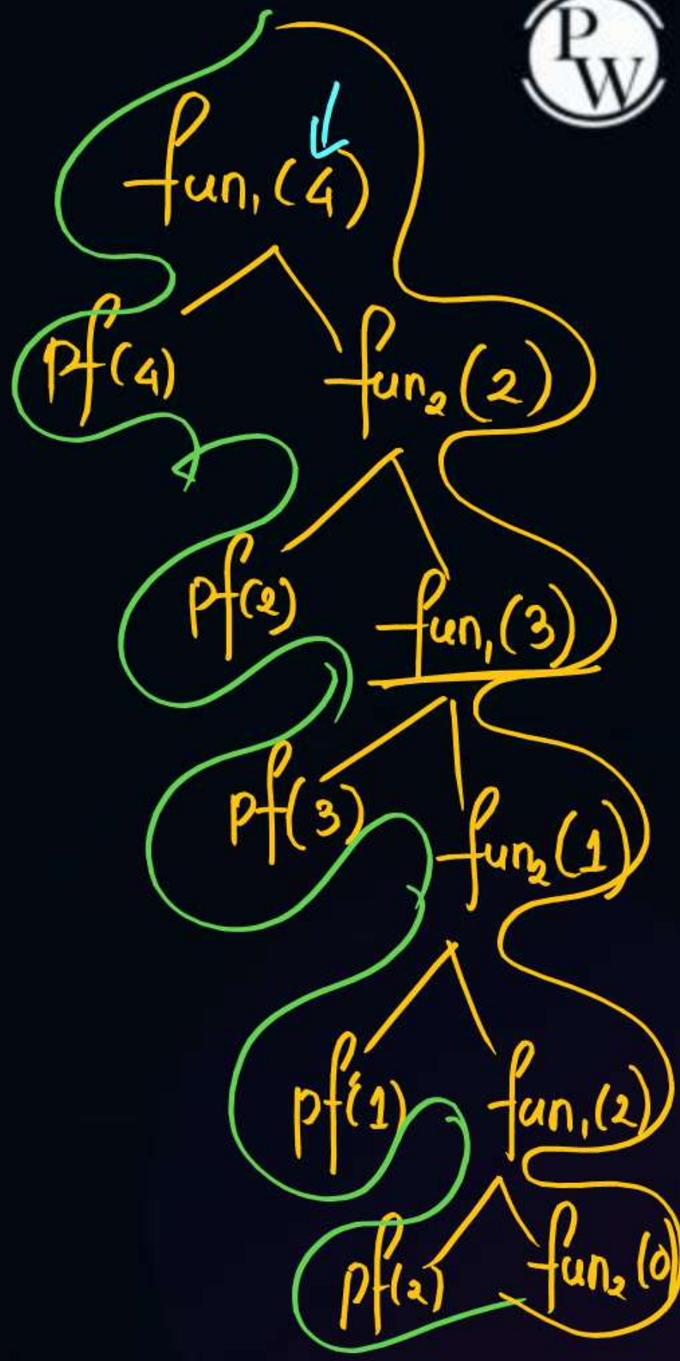
printf ("%d" , n);

fun1(++n) ;

}
```

3, 1, 2

4, 2, 3, 1, 2





Indirect Recursion

```
void fun1 (int n) {

if (n = =0 ) return;

printf ("%d" , n);

fun2 (n - 2);

}

int main(){

fun1(3);

}
```

```
void fun2 (int n) {

if (n = = 0) return ;

printf ("%d" , n);

fun1(++n) ;

}
```



Q. if $fun_2(3)$ is called then total No. of times printf statement executed [6]



Indirect Recursion

3, 4, 2, 3, 1, 2

```
void fun1 (int n) {

if (n = =0 ) return;

printf ("%d" , n);

fun2 (n - 2);

}

int main(){

fun1(3);

}
```

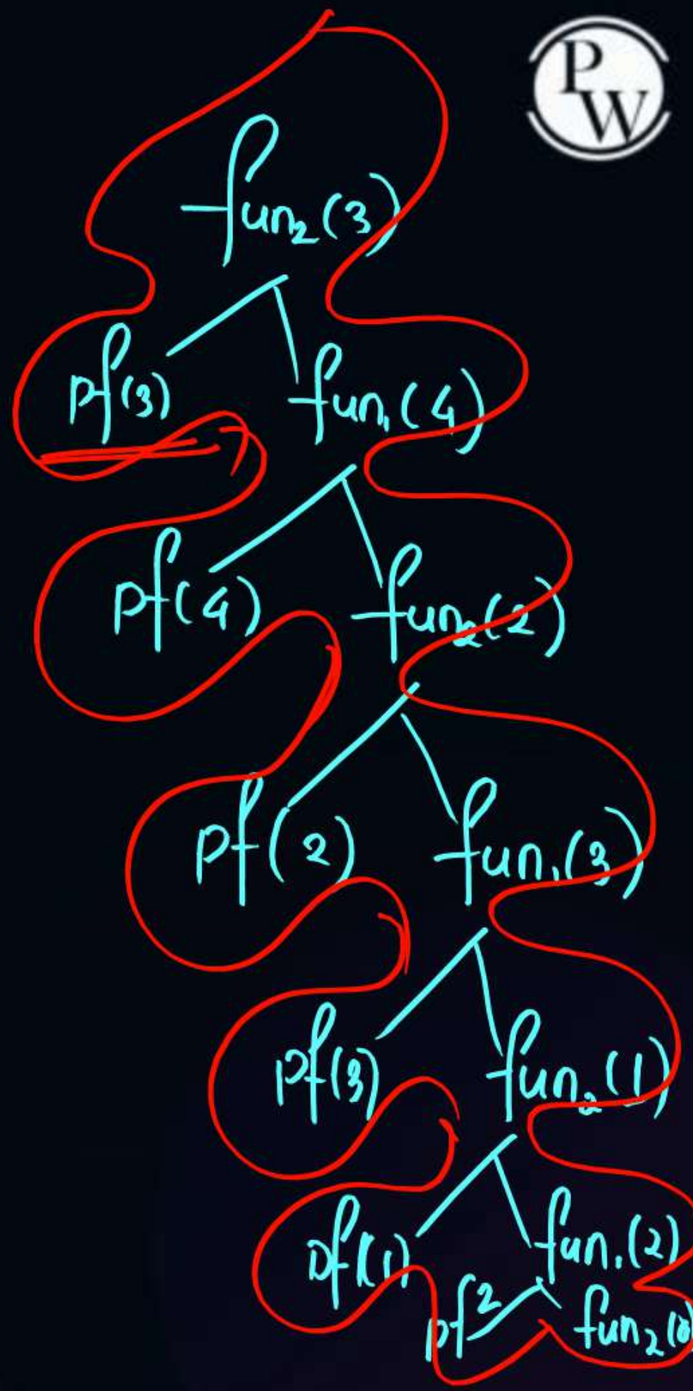
```
void fun2 (int n) {

if (n = = 0) return ;

printf ("%d" , n);

fun1(++n) ;

}
```





Indirect Recursion

```
void fun1 (int n) {

if (n = =0 ) return;

printf ("%d" , n);

fun2 (n - 2);

}

int main(){

fun1(3);

}
```

```
void fun2 (int n) {

if (n = = 0) return ;

printf ("%d" , n);

fun1(++n) ;

printf("%d", n);

}
```

if fun1(4) is called then output of program is



Indirect Recursion

```
void fun1 (int n) {

if (n = =0 ) return;

printf ("%d" , n);

fun2 (n - 2);

}

int main(){

fun1(3);

}
```

```
void fun2 (int n) {

if (n = = 0) return ;

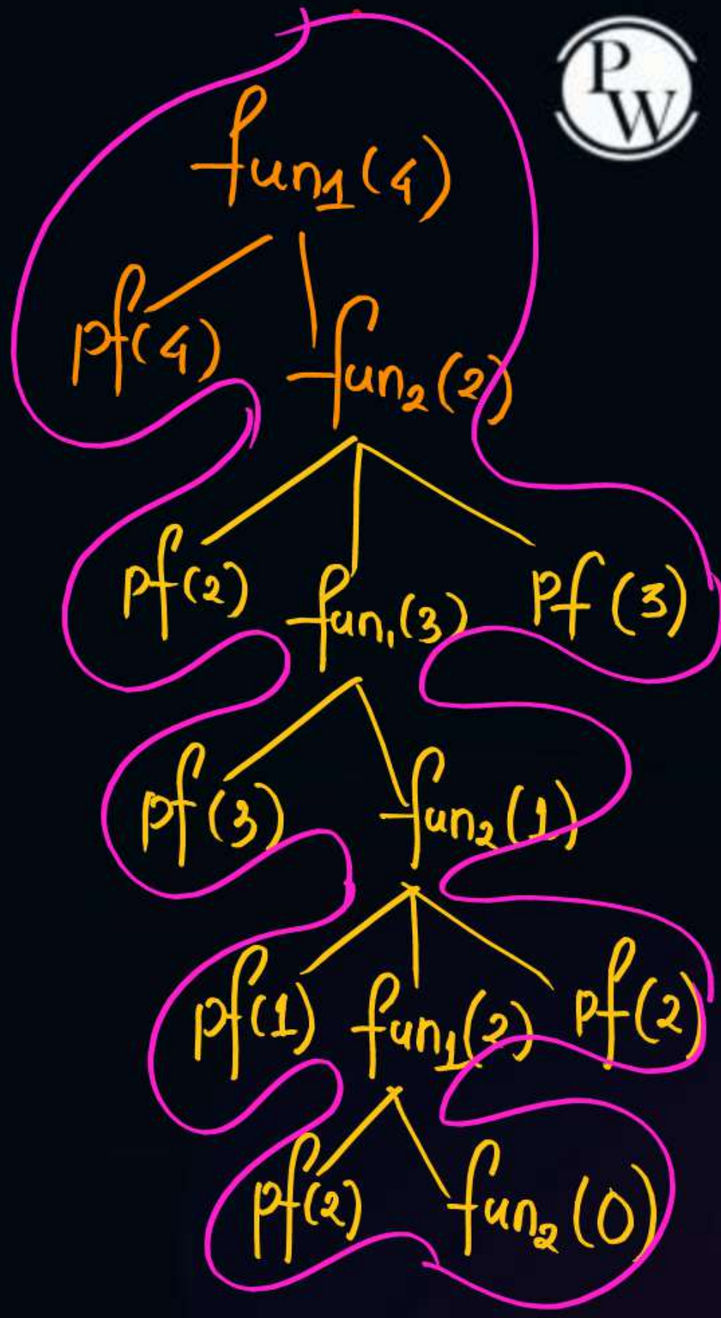
printf ("%d" , n);

fun1(++n) ;

printf("%d", n);

}
```

4, 2, 3, 1, 2, 2, 3





Question

#Q. Consider the following program

```
int fun(int x){
    if(x>3)
        return fun(x-4) + fun(x-1) + 1;
    return 1;
}
```

Find the value returned by fun(5)



Ackermann function

$$A(x,y) = \begin{cases} y+1 & if\ x = 0 \\ A(x-1,1) & if\ y = 0 \\ A(x-1, A(x,y-1)) & otherwise \end{cases}$$

HW

A(1,2) =



Question

What is the output printed by the following program?

```
# include <stdio.h>
int f(int n, int k) {
if (n == 0) return 0;
else
if (n%2)
  return f(n/2, 2*k) + k;
else
  return f(n/2, 2*k) - k;
}
```

```
int main () {
printf("%d",f(20,1));

return 0;
     }
```

(A) 5

(B) 8

(C) 9

(D) 20



2 mins Summary

Topic — Nested Recursion

Topic — Indirect Recursion

Topic

Topic

Topic

THANK - YOU

